

A.3. Supplementary Information to Chapter 4

A.3.1. Preliminary Force Field Parameters

The preliminary version differs from the published parameters [150] only by the parameters for Cl–Cl interactions. The preliminary parameters are contributed by Leonid Komissarov.

40-2 > 'C.C.O.Cl', 'C.O.Cl'

41 ! Number of general parameters

50.0000

9.5469

26.5405

0.6863

2.7295

70.0000

1.0588

4.1262

12.1176

13.3056

-68.9784

0.0000

10.0000

0.0000

33.8667

6.0891

1.0563

2.0384

6.1431

6.9290

0.3989

3.9954

0.0000

5.7796

10.0000

1.9487

0.0000
2.1645
1.5591
0.1000
2.1365
0.6991
50.0000
1.8512
0.0000
0.0000
0.0000
1.0000
2.6962
0.0000
0.0000

4 ! Number of atoms

C 1.3674 4.0000 12.0000 2.0453 0.1444 0.9500 1.1706 4.0000
9.0000 1.5000 4.0000 27.5134 79.5548 5.0191 7.0500 0.0000
1.1168 0.0000 181.0000 14.2732 24.4406 6.7313 0.8563 0.0000
-4.1021 5.0000 1.0564 4.0000 2.9663 0.0000 0.0000 0.0000
H 0.9479 1.0000 1.0080 1.1364 0.0232 0.9900 -0.1000 1.0000
9.0643 4.7746 1.0000 0.0000 121.1250 4.7757 9.7732 1.0000
-0.1000 0.0000 62.4879 2.5194 2.3785 0.2223 1.0698 0.0000
-15.7683 2.1488 1.0338 1.0000 2.8793 0.0000 0.0000 0.0000
O 1.1939 2.0000 15.9990 1.9289 0.1201 0.9900 1.0981 6.0000
10.4842 8.2916 4.0000 28.8967 116.0768 7.9703 7.0500 2.0000

1.0479 20.0000 60.8726 10.0338 2.2024 0.9942 0.9745 0.0000
-3.6141 2.7025 1.0493 4.0000 2.9225 0.0000 0.0000 0.0000
CI 1.4622 1.0000 35.4500 2.1699 0.1143 0.5854 -1.0000 7.0000
9.4263 6.2402 1.0000 2.3692 0.0000 8.7026 18.3815 2.0000
-1.0000 5.0303 28.0000 10.5240 4.0008 0.2045 0.8656 0.0000
-11.9991 2.6929 1.0338 4.0000 2.0389 0.0000 0.0000 0.0000
10 ! Number of bonds

1 1 80.8865 107.9944 52.0636 0.5218 -0.3636 1.0000 34.9876 0.7769
6.1244 -0.1693 8.0804 1.0000 -0.0586 8.1850 1.0000 0.0000
1 2 179.5195 0.0000 0.0000 -0.5242 0.0000 1.0000 6.0000 0.7187
5.4740 1.0000 0.0000 1.0000 -0.1144 6.7029 0.0000 0.0000
2 2 113.9232 0.0000 0.0000 -0.5971 0.0000 1.0000 6.0000 0.9093
1.7152 1.0000 0.0000 1.0000 -0.0450 6.0710 0.0000 0.0000
1 3 136.4945 164.1201 5.5000 -0.9159 -0.1075 1.0000 10.6519 0.8644
0.6858 -0.4602 9.5754 1.0000 -0.1745 4.5987 0.0000 0.0000
3 3 148.0798 155.2406 20.1160 -1.0000 -0.1254 1.0000 33.0027 0.7790
0.7673 -0.1697 7.0028 1.0000 -0.1300 5.1959 1.0000 0.0000
2 3 169.1351 0.0000 0.0000 -0.8810 0.0000 1.0000 6.0000 0.5757
1.5482 1.0000 0.0000 1.0000 -0.1788 4.6622 0.0000 0.0000
1 4 110.2423 0.0000 0.0000 0.2779 0.0000 1.0535 6.0000 0.1394
18.0983 -0.2500 0.0000 1.0000 -0.4777 3.2360 0.0000 0.0000
2 4 148.8040 0.0000 0.0000 -0.2898 0.0000 0.8642 6.0000 1.7483
2.6898 -0.2000 0.0000 1.0000 -0.1352 3.6144 0.0000 0.0000
3 4 68.6034 0.0000 0.0000 -0.9994 0.0000 0.0051 6.0000 0.8167
2.7959 -0.2000 0.0000 1.0000 -0.1428 8.9863 0.0000 0.0000
4 4 91.7101 0.0000 0.0000 0.0713 0.0000 0.4976 6.0000 4.9999
-0.0523 -0.2500 0.0000 1.0000 -0.0100 7.2315 0.0000 0.0000

6 ! Number of off-diagonal terms

1 2 0.1253 1.5717 9.9736 1.2057 -1.0000 -1.0000

2 3 0.1125 1.6311 8.7528 1.0929 -1.0000 -1.0000

1 3 0.0953 1.7397 8.8986 1.4256 1.1067 1.1265

1 4 0.2182 1.4549 12.1783 1.8100 -1.0000 -1.0000

2 4 0.2059 0.9935 12.8651 1.2900 -1.0000 -1.0000

3 4 0.2175 1.1010 15.9842 1.7300 -1.0000 -1.0000

29 ! Number of angles

1 1 1 76.1370 34.6920 1.1328 0.0000 0.0050 0.3556 1.8065

1 1 2 68.0572 9.9461 4.7000 0.0000 0.4566 0.0000 1.8532

2 1 2 65.6815 35.0000 1.8622 0.0000 0.0490 0.0000 1.0937

1 2 2 0.0000 4.0000 7.2043 0.0000 0.0000 0.0000 1.0728

1 2 1 0.0000 3.4110 7.7350 0.0000 0.0000 0.0000 1.0400

2 2 2 0.0000 30.0000 5.6235 0.0000 0.0000 0.0000 1.0400

1 1 3 78.3624 13.0773 9.0480 0.0000 0.1270 52.1129 2.3964

3 1 3 76.7101 24.3833 5.8613 -21.8559 2.6395 -32.6534 3.6179

2 1 3 79.1288 30.0000 1.4632 0.0000 0.2065 0.0000 2.0000

1 3 1 80.7352 16.4130 4.9987 0.0000 0.0843 0.0000 1.0137

1 3 3 85.4436 14.4937 3.9928 0.0000 1.4350 44.5320 1.1348

3 3 3 89.9282 32.1199 2.7181 0.0000 0.3323 57.6122 1.0000

1 3 2 82.9640 32.4874 0.8777 0.0000 0.9627 0.0000 1.0010

2 3 3 85.7838 17.3139 1.9157 0.0000 3.6306 0.0000 2.1858

2 3 2 84.2527 33.1226 0.6730 0.0000 0.7238 0.0000 2.4348

1 2 3 0.0000 14.4588 3.1507 0.0000 3.4571 0.0000 1.0149

3 2 3 0.0000 0.9696 3.6303 0.0000 0.0000 0.0000 1.6987

2 2 3 0.0000 0.5797 1.9739 0.0000 0.0000 0.0000 2.4494

1 4 4 0.0000 97.9606 8.0433 0.0000 0.0000 0.0000 3.8708

1 1 4 55.8900 20.7732 1.7644 0.0000 4.8904 0.0000 1.6417

1 3 4 2.6434 -9.1685 9.0474 0.0000 1.5171 0.0000 6.1281
 2 1 4 75.5068 -0.5924 9.9246 0.0000 4.7767 0.0000 5.8248
 2 3 4 126.8583 41.5246 0.8000 0.0000 0.5282 0.0000 1.2442
 3 1 4 61.8784 72.7241 0.3156 0.0000 4.9506 0.0000 3.4766
 3 2 4 0.0000 91.4780 4.2256 0.0000 0.0000 0.0000 1.5737
 3 3 4 73.5892 20.0466 1.1863 0.0000 4.9940 0.0000 4.8561
 4 1 4 51.0157 39.4737 0.8047 0.0000 2.1395 0.0000 1.1603
 4 3 4 10.9340 -29.7636 1.4274 0.0000 1.2533 0.0000 7.9693
 4 4 4 0.0000 42.8758 4.2241 0.0000 0.0000 0.0000 1.6690

32 ! Number of torsions

1 1 1 1 2.0474 32.6719 0.5282 -9.0000 -2.6449 0.0000 0.0000
 1 1 1 2 1.6328 78.4995 -0.1514 -6.9161 -2.9986 0.0000 0.0000
 2 1 1 2 2.4142 78.7025 0.3506 -8.8640 -6.9283 0.0000 0.0000
 1 1 1 3 -0.7104 22.6038 0.5309 -2.0000 -0.6614 0.0000 0.0000
 2 1 1 3 1.9323 52.9368 0.6554 -8.8118 -3.9854 0.0000 0.0000
 3 1 1 3 -1.2500 1.1248 -0.1230 -9.9453 -3.9000 0.0000 0.0000
 1 1 3 1 -0.6848 56.7751 -1.2733 -2.2937 -4.0000 0.0000 0.0000
 1 1 3 2 -1.4557 78.6279 0.9945 -3.2742 -2.4240 0.0000 0.0000
 2 1 3 1 0.6928 78.1546 0.5608 -3.1713 -3.7301 0.0000 0.0000
 2 1 3 2 -1.4343 77.0699 0.9875 -3.4139 -1.4053 0.0000 0.0000
 1 1 3 3 0.5153 2.1584 0.2000 -6.5859 -3.0000 0.0000 0.0000
 2 1 3 3 0.2018 80.0000 0.3778 -2.5000 -2.8750 0.0000 0.0000
 3 1 3 1 -1.9875 79.2591 1.0000 -2.4206 -3.9342 0.0000 0.0000
 3 1 3 2 -1.1000 78.8002 -1.0000 -2.6282 -4.0000 0.0000 0.0000
 3 1 3 3 -1.0000 83.5323 4.3660 -2.6805 -1.2938 0.0000 0.0000
 1 3 3 1 3.4682 0.0781 0.9887 -6.1195 -0.5004 0.0000 0.0000
 1 3 3 2 1.0000 16.5478 -1.0313 -2.0000 -2.6888 0.0000 0.0000
 2 3 3 2 4.0818 -3.2744 -0.9664 -7.1634 -3.0000 0.0000 0.0000

1 3 3 3 4.2014 -10.0642 1.8690 -2.4805 -2.5000 0.0000 0.0000
2 3 3 3 1.0000 -10.0500 -1.0000 -2.1946 -0.5300 0.0000 0.0000
3 3 3 3 1.0000 1.6871 3.0000 -6.2660 -0.5500 0.0000 0.0000
1 1 1 4 4.1485 38.1222 2.2912 -10.8271 19.5577 0.0000 0.0000
1 3 1 4 13.5963 143.0497 -0.6551 -10.3770 50.3494 0.0000 0.0000
1 1 3 4 4.6153 -41.2146 2.6095 -2.3656 -59.9900 0.0000 0.0000
1 3 3 4 24.0662 -13.6312 -9.8252 1.2433 -59.9340 0.0000 0.0000
2 1 1 4 -2.5298 120.3131 0.8292 -5.3840 19.6104 0.0000 0.0000
2 1 3 4 3.1162 64.2411 1.3957 -4.6578 -59.4257 0.0000 0.0000
2 3 1 4 0.8541 -25.0788 -0.8283 -3.7322 36.5262 0.0000 0.0000
2 3 3 4 6.0246 29.4544 -0.4845 -6.3106 47.2012 0.0000 0.0000
3 1 1 4 -1.8658 158.9111 1.1132 -7.2577 -0.4250 0.0000 0.0000
3 1 3 4 2.7912 54.0974 -2.1091 -2.9448 -58.0874 0.0000 0.0000
4 1 1 4 -13.2905 144.9389 3.9884 -4.2634 -30.4780 0.0000 0.0000
2 ! Number of hydrogen bonds
3 2 3 1.8130 -3.5409 2.3815 21.9463
4 2 3 1.8319 -7.9316 3.2138 3.2323